

ECA04-ANALOG CIRCUITS

LIST OF EXPERIMENTS

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Exp.No. 1 Design of Voltage Series feedback amplifiers using BJT

AIM: To construct an voltage series circuit and Measure the gain , Plot its input and output waveforms

OBJECTIVES

On completion of the experiment students will be able to

- ☐ Assemble an voltage-series circuit
- ☐ Test the dc condition and ascertain the working condition of the amplifier
- ☐ Understand the characteristics of voltage-series circuit

EQUIPMENTS / COMPONENTS

Sl. no.	Name and Specification	Quantity
1.	Resistors 1 K Ω	1 no.
	2.2 K Ω	1 no.
	10 K Ω	1 no.
	22 K Ω	1 no.
	33 K Ω	1 no.
2.	Capacitor 1 μ F	2Nos
3.	Transistor BC 107	1 no.
4.	Function generator 0 to 1 MHz	1 no.
5.	Oscilloscope 0 to 20 MHz	1 no.
6.	Multimeter	1 no.
7.	Breadboard	1 no.
8.	Connecting wires	

PRINCIPLE

Voltage – series in emitter follower is the popular name for common collector amplifier. Its voltage gain is approximately unity (without R_L voltage gain is unity). It has high input impedance and low output impedance. Thus emitter follower has less loading effect and is suitable for impedance matching.

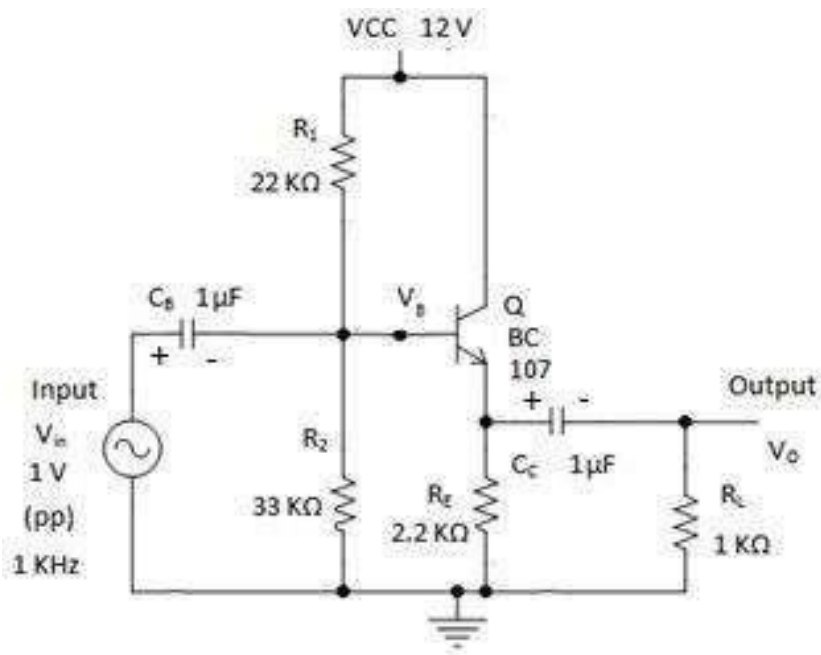
Since collector is directly connected to dc source, it appears to be grounded for ac signal. Output is taken from the emitter terminal. The output voltage is in phase and is equal to the input signal. Since the amplitude and phase of the output (emitter) follows the input (base), the circuit is called emitter

follower. In this circuit voltage divider biasing is used for base bias. R_E acts as the load for signal at the output circuit. R_E also provides a negative feedback in the circuit.

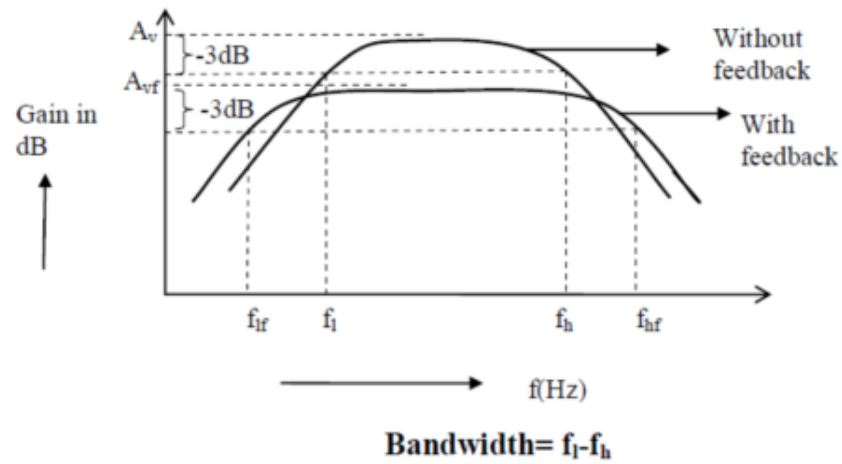
PROCEDURE

- 1) Test the components
- 2) Assemble the circuit
- 3) Measure the dc condition using multimeter and verify whether the transistor is in active region
- 4) Apply 1Vpp, 1 KHz sinusoidal signal as input
- 5) Observe the voltages at input point (V_{in}), at base, at emitter and at the output point (V_o) without R_L
- 6) Measure the amplitudes and dc levels
- 7) Plot the waveforms
- 8) Observe and measure V_o with $R_L = 10\text{ K}\Omega$ and $R_L = 1\text{ K}\Omega$
- 9) Calculate the voltage gain for the above three conditions of R_L

CIRCUIT DIAGRAM



MODEL GRAPH :



TABULATION 1

Measurement of frequency response of voltage series feedback amplifiers .

$V_{in} =$

Frequency (in Hz)	V_o (Volts)	Gain= V_o/V_{in}	Gain (dB) = $20 \log(V_o/V_{in})$

RESULT

1. The Voltage series feedback amplifier was designed, constructed and its

frequency response was plotted.

2. The following parameters were observed.

Frequency response data	Voltage series feedback amplifier	Amplifier without feedback
Mid- band Voltage gain		
Bandwidth		

Case.1.1.

Consider a single-stage common-emitter amplifier that provides a voltage gain of 30 without feedback. A portion of the output voltage is fed back in series with the input through a feedback network.

Given: Open-loop voltage gain (A) = 30, Closed-loop voltage gain (A_f) = 10, and Input resistance (R_{in}) = 2 k Ω .

Design the feedback network and calculate:

- (i) The feedback factor (β).
- (ii) The loop gain ($A\beta$).
- (iii) The percentage reduction in gain due to feedback.
- (iv) The new input resistance ($R_{in,f}$).

Case.1.2.

A voltage series feedback amplifier has the following parameters:

Open-loop voltage gain (A) = 50, Feedback factor (β) = 0.02, Input resistance (R_{in}) = 1 k Ω , and Output resistance (R_{out}) = 100 Ω .

Calculate:

- (i) The closed-loop gain (A_f).
- (ii) The input resistance with feedback ($R_{in,f}$).
- (iii) The output resistance with feedback ($R_{out,f}$).
- (iv) The improvement in bandwidth if the open-loop bandwidth is 20 kHz.

Case.1.3.

Design a voltage series feedback amplifier to achieve stable gain and improved input impedance.

Given: Open-loop gain (A) = 100, Desired closed-loop gain (A_f) = 20, Source resistance (R_s) = 600 Ω , and Load resistance (R_L) = 2.2 k Ω .

Determine:

- (i) The required feedback factor (β).
- (ii) The loop gain ($A\beta$).
- (iii) The minimum open-loop input resistance (R_{in}) needed to achieve a closed-loop input resistance of 10 k Ω .
- (iv) Suitable resistor values for the feedback network (R_1 and R_2) to realize the required β .

Exp No. 2 DESIGN OF VOLTAGE SHUNT FEEDBACK AMPLIFIER USING BJT

AIM:

To obtain the frequency response of a voltage shunt feedback (with and without) amplifier and obtain the bandwidth.

OBJECTIVES

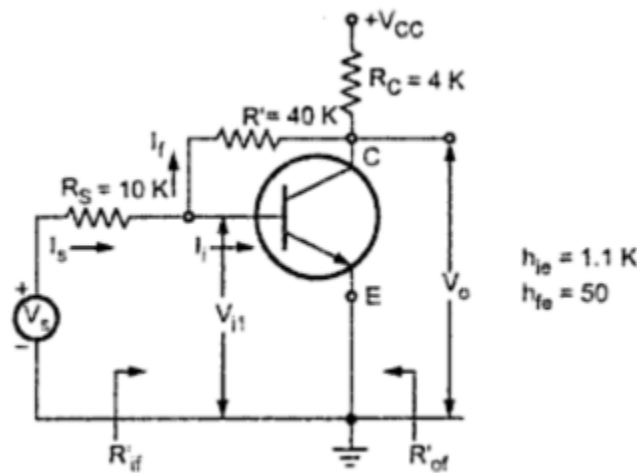
On completion of the experiment students will be able to

- ☐ Assemble an voltage shunt circuit
- ☐ Test the dc condition and ascertain the working condition of the amplifier

EQUIPMENTS / COMPONENTS

Sl. no.	Name and Specification	Quantity
1.	Resistors 4 K Ω	1 no.
	2.2 K Ω	1 no.
	40 K Ω	1 no.
	10 K Ω	1 no.
	33 K Ω	1 no.
2.	Capacitor 1 μ F	2Nos
3.	Transistor BC 107	1 no.
4.	Function generator 0 to 1 MHz	1 no.
5.	Oscilloscope 0 to 20 MHz	1 no.
6.	Multimeter	1 no.
7.	Breadboard	1 no.
8.	Connecting wires	

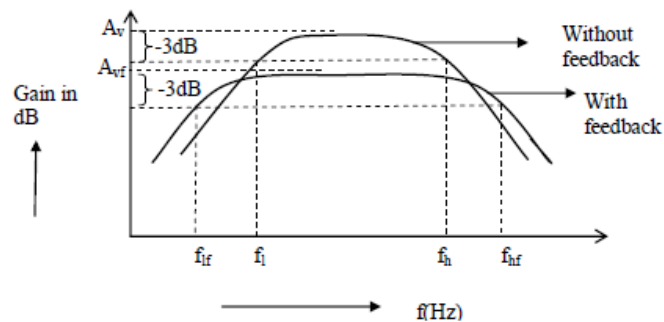
CIRCUIT DIAGRAMS



PROCEDURE:

1. Connections are made as per the circuit diagram.
2. A 10V DC supply is given to the circuit for biasing.
3. The circuit is connected without feedback i.e., without R_F
4. A certain amplitude of input signal (say 20mV) is kept constant using the function generator and for different frequencies the output voltage from CRO are noted.
5. Now, the circuit is connected with feedback i.e., with R_F .
6. By keeping the input signal constant the output voltages for different frequencies are noted from CRO.
7. Gain with and without feedback is calculated from the formula

MODEL GRAPH :



$$\text{Bandwidth} = f_1 - f_h$$

TABULATION 1

Measurement of frequency response of voltage shunt feedback amplifiers

V_{in} = -----

Frequency (in Hz)	V_o (Volts)	Gain = V_o/V_{in}	Gain (dB) = $20 \log(V_o/V_{in})$

RESULT

1. The voltage shunt feedback amplifier was designed constructed and its frequency response was plotted.
2. The following parameters were observed.

Frequency response data	Voltage shunt feedback amplifier	Amplifier without feedback
Mid- band Voltage gain		
Bandwidth		

Case.2.1.

A single-stage Voltage shunt common-emitter amplifier provides an open-loop voltage gain of 200. A portion of the output voltage is fed back in parallel with the input using a resistive network.

Given: $A = 200$, $\beta = 0.01$, Input resistance (R_{in}) = 3 k Ω , Output resistance (R_{out}) = 500 Ω .

Calculate:

- (i) The closed-loop voltage gain (A_f).
- (ii) The new input resistance ($R_{in,f}$).
- (iii) The new output resistance ($R_{out,f}$).
- (iv) The percentage reduction in gain due to feedback.

Case.2.2.

A voltage shunt feedback amplifier is used for current sensing applications.

Given: Open-loop gain (A) = 150, Feedback factor (β) = 0.02, Open-loop output resistance (R_{out}) = 1 k Ω , and Load resistance (R_L) = 2.2 k Ω .

Determine:

- (i) The closed-loop gain (A_f).
- (ii) The effective output resistance with feedback ($R_{out,f}$).
- (iii) The improvement in linearity and distortion due to feedback.
- (iv) The effect of feedback on input resistance.

Case.2.3.

Design a voltage shunt feedback amplifier to achieve high output stability and low input impedance.

Given: Open-loop gain (A) = 100, Desired closed-loop gain (A_f) = 25, and Open-loop input resistance (R_{in}) = 4 k Ω .

Find:

- (i) The required feedback factor (β).
- (ii) The loop gain ($A\beta$).
- (iii) The new input resistance with feedback ($R_{in,f}$).
- (iv) Suggest a feedback resistor network (R_f and R_1) to realize β .

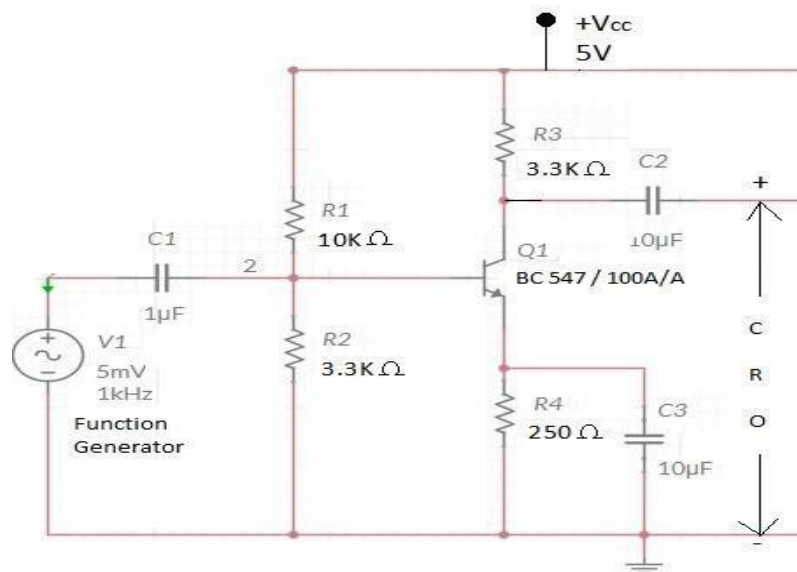
3. FREQUENCY RESPONSE OF COMMON EMITTER AMPLIFIER CIRCUIT USINGBJT

Aim: To Plot the Frequency Response Characteristics of Common Emitter amplifier circuit using BJT.

Apparatus Required:

S.No.	Component	Specification	Quantity
1.	Transistor	BC547	1
2.	Resistor	3.3k Ω ,10k Ω ,250 Ω	2,1
3.	Capacitor	10 μ F, 1 μ F	2,1
4.	Function Generator	(0-3MHz)	1
5.	Regulatedpower supply	(0-30)V	1
6.	Cathode Ray Oscilloscope	(0-30MHz)	1
7.	BreadBoard	-	1
8.	ConnectingWires	-	-

Circuit diagram:

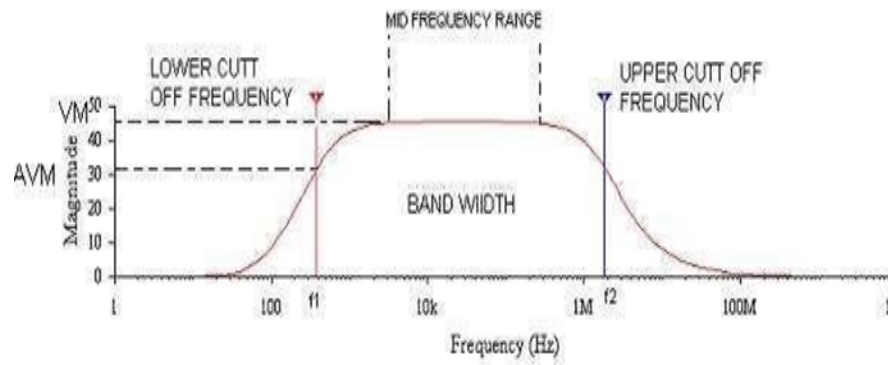


Procedure for Common Emitter Amplifier Circuit using BJT:

1. Connect the circuit as per the circuit diagram which is shown above.
2. Initially connected the probe across the function generator as per shown in the circuit diagram to set the input signal.

[illegible]

Model Graph:



Result:

Common Emitter Amplifier circuit was designed and bandwidth calculated.

Bandwidth = _____(KHz)

Case.3.1.

Design a common emitter amplifier circuit with the following specifications:

- Voltage gain (A_v) = 50
- Input resistance (R_{in}) = 1 k Ω
- Load resistance (R_L) = 5 k Ω
- Power supply voltage (V_{cc}) = 12V
- β (current gain) of the BJT = 100

Case.3.2.

Design a common emitter amplifier circuit with a BJT transistor having a DC current gain (β) of 100. The amplifier should have a voltage gain of 50 and a collector current (I_C) of 2 mA. The power supply voltage (V_{CC}) is 10 volts. Assume the transistor has a base-emitter voltage drop (V_{BE}) of 0.7 volts and a collector-emitter voltage drop (V_{CE}) of 0.2 volts in the active region. Design the circuit and calculate the values of resistance components.

Case.3.3.

Design a common emitter amplifier circuit using a BJT transistor with the following specifications:

- Desired voltage gain (A_v) = 50
- Input resistance (R_{in}) = 1 k Ω
- Output resistance (R_{out}) = 10 k Ω
- Supply voltage (V_{cc}) = 12 V
- Transistor beta (β) = 100
- Maximum collector current (I_c) = 10 mA

Case.3.4.

Design a common emitter amplifier circuit with a voltage gain of 50. The collector current is 2 mA, and the supply voltage is 10 V. The transistor has a beta (β) of 100.

4. DESIGN OF RC PHASE SHIFT OSCILLATOR AND LC OSCILLATOR CIRCUIT USING BJT

Aim:

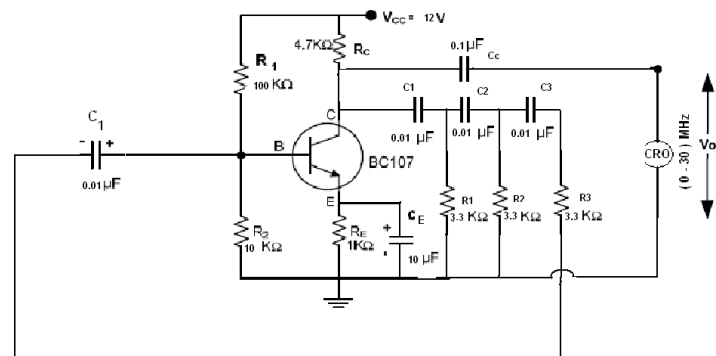
To setup RC and LC oscillator and

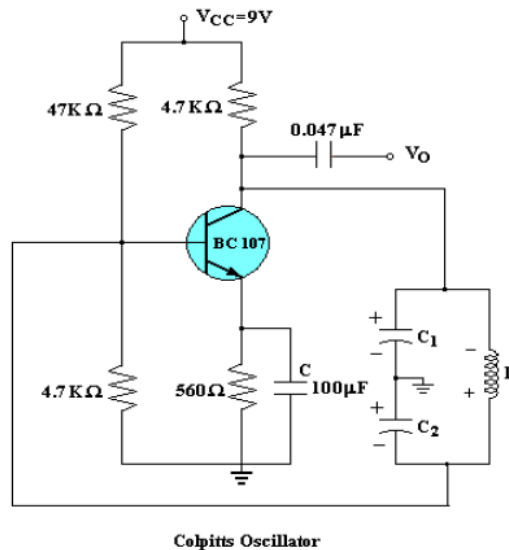
- (i) Plot the output waveform
- (ii) Measure the frequency of oscillation

Apparatus Required:

S.No.	Component	Specification	Quantity
1.	Transistor	BC 107	1
2.	Resistor	4.7k Ω ,10 k Ω 100k Ω ,3.3K Ω ,1 k Ω	Each 1
3.	Capacitor	0.1 μ F, 10 μ F,0.01 μ F,	1,3
4.	Transistor	BF195	1
5.	Resistor	1.8k Ω ,47k Ω 560k Ω ,10k Ω	Each 1
6.	Capacitor	1 μ F,0.01 μ F,	Each 1
7.	Inductor	20mh	1
8.	Regulatedpower supply	(0-30)V	1
9.	Cathode Ray Oscilloscope	(0-30MHz)	1
10.	BreadBoard	-	1
11.	ConnectingWires	-	-

Circuitdiagram:





Procedure:

1. Connections are made as per circuit diagram.
2. Connect CRO output terminals and observe the waveform.
3. Calculate practically the frequency of oscillations by using the expression $f = 1 / T$ (T = Time period of the waveform).
4. Compare the values of oscillations both theoretically and practically..

Formula:

Frequency of oscillation is given by:

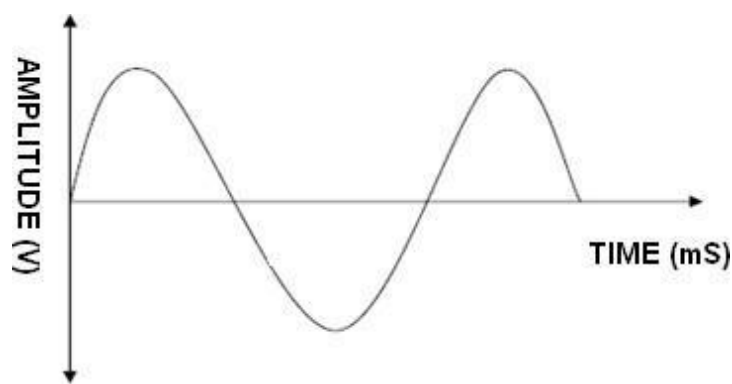
$$f = \frac{1}{2\pi RC\sqrt{6}} \quad -RC$$

$$f = \frac{1}{2\pi(L C_{eq})^{1/2}} \quad \text{where} \quad C_{eq} = \frac{C_1 C_2}{(C_1 + C_2)}$$

Tabulation:

Oscillator	AMPLITUDE(V)	TIMEPERIOD(ms)	FREQUENCY(Hz)
RC			
LC			

Model Graph:



Result:

RC Phase Shift Oscillator circuit and LC Hartely oscillator was designed and practical frequency calculated.

Frequency $f =$ _____(Hz)

Case.4.1.

Design an RC phase shift oscillator using a BJT transistor with the following specifications:

Frequency of oscillation (f_o) = 1 kHz

Supply voltage (V_{cc}) = 12V

The gain of the transistor (β) = 100

Case.4.2.

Design an RC phase shift oscillator using a BJT with a frequency of oscillation at 1 kHz. Assume the transistor has a gain (β) of 100.

Case.4.3.

Design a Colpitts oscillator using a BJT with the following specifications:

Desired oscillation frequency: 1 MHz

Supply voltage: 9 V

Desired output voltage swing: 2 V peak-to-peak

Transistor: BC547 (common NPN transistor)

Case.4.4.

Consider the following circuit for a Hartley oscillator using a BJT:

Given:

$L_1 = 100 \mu\text{H}$ and C_1 and $C_2 = 100 \text{ pF}$ (β (current gain of the BJT) = 100

V_{cc} (supply voltage) = 12V V_{be} (base-emitter voltage) = 0.7V

Find the oscillation frequency and the component values.

5. DESIGN OF ASTABLE MULTIVIBRATOR USING BJT

Aim:

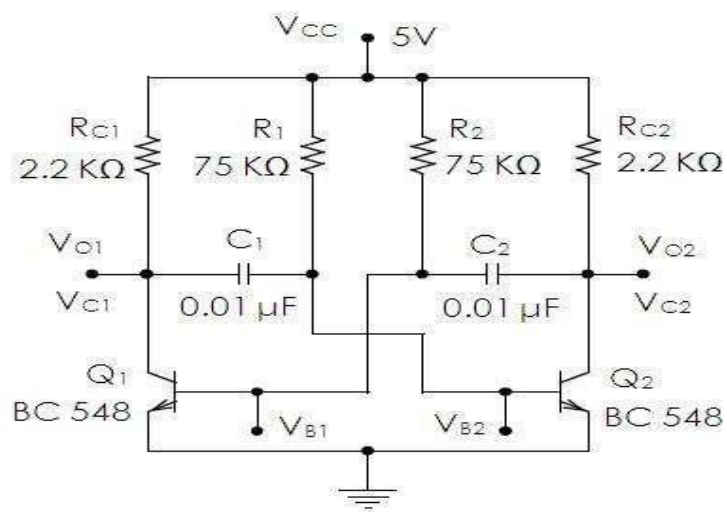
To construct a **transistor-based Astable Multivibrator** circuit for **1 kHz** and:

- (i) Plot the **collector** and **base** waveforms.
- (ii) Measure the **frequency of oscillation**.

Apparatus Required:

S.No.	Component	Specification	Quantity
1.	Transistor	BC548	2
2.	Resistor	2.2k Ω , 75 k Ω	Each 2
3.	Capacitor	0.01 μ F	2
4.	Regulated power supply	(0-30)V	1
5.	Cathode Ray Oscilloscope	(0-30MHz)	1
6.	BreadBoard	-	1
7.	Connecting Wires	-	-

Circuit diagram:



Procedure:

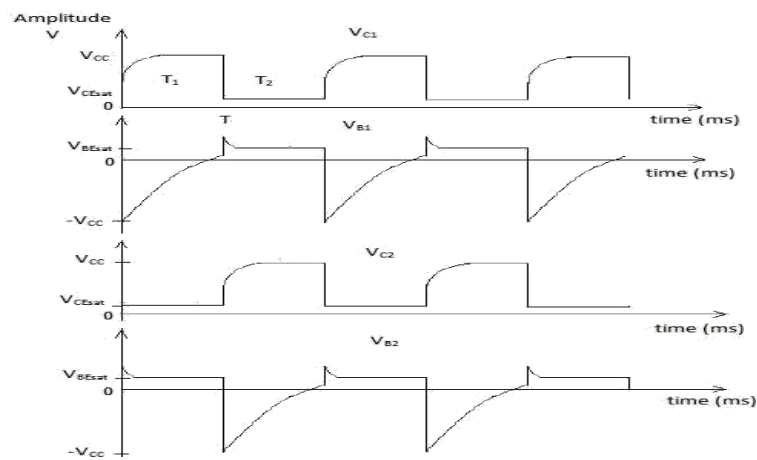
- 1. Test the components.
- 2. Assemble the circuit on a breadboard.
- 3. Switch ON the power supply.
- 4. Connect the outputs of the circuit to an oscilloscope.
- 5. Observe the **collector** and **base** waveforms of the two transistors.
- 6. Measure the **frequency** and **amplitude** of the outputs.
- 7. Plot the waveforms.

Formula:

Time Period $T=T_1+T_2$ where T_1 = ON time T_2 = OFF time

Frequency $f=1/ T$ in Hz.

Model Graph:



Tabulation:

S.no	Amplitude (involts)	TON (ms)	TOFF (ms)	Frequency (inHz)
Vo1				
V02				

Result:

Astable Multivibrator was designed and frequency of oscillation was calculated.

Frequency of oscillation $f = \underline{\hspace{2cm}}$ (Hz)

Case.5.1.

Design an astablemultivibrator circuit using two NPN bipolar junction transistors (BJTs), such as 2N3904, to generate square wave oscillations with a frequency of 1 kHz. Supply voltage V_{cc} is 12V.

Case.5.2.

Design an astablemultivibrator circuit using BJTs that operates at a frequency of 1 kHz with a duty cycle of 50%. Use standard silicon BJTs with a supply voltage of 9V.

Case.5.3.

Design an astablemultivibrator circuit using two NPN BJTs to produce a square wave output with a frequency of 1 kHz. Use resistors and capacitors to achieve the desired frequency. Supply voltage is 9V.

Case.5.4.

Design an astablemultivibrator circuit using two NPN bipolar junction transistors (BJTs) to generate a square wave output with a frequency of 1 kHz. The circuit should operate from a single DC power supply of 12V.

6. Simulation of CB amplifier using LTSpice

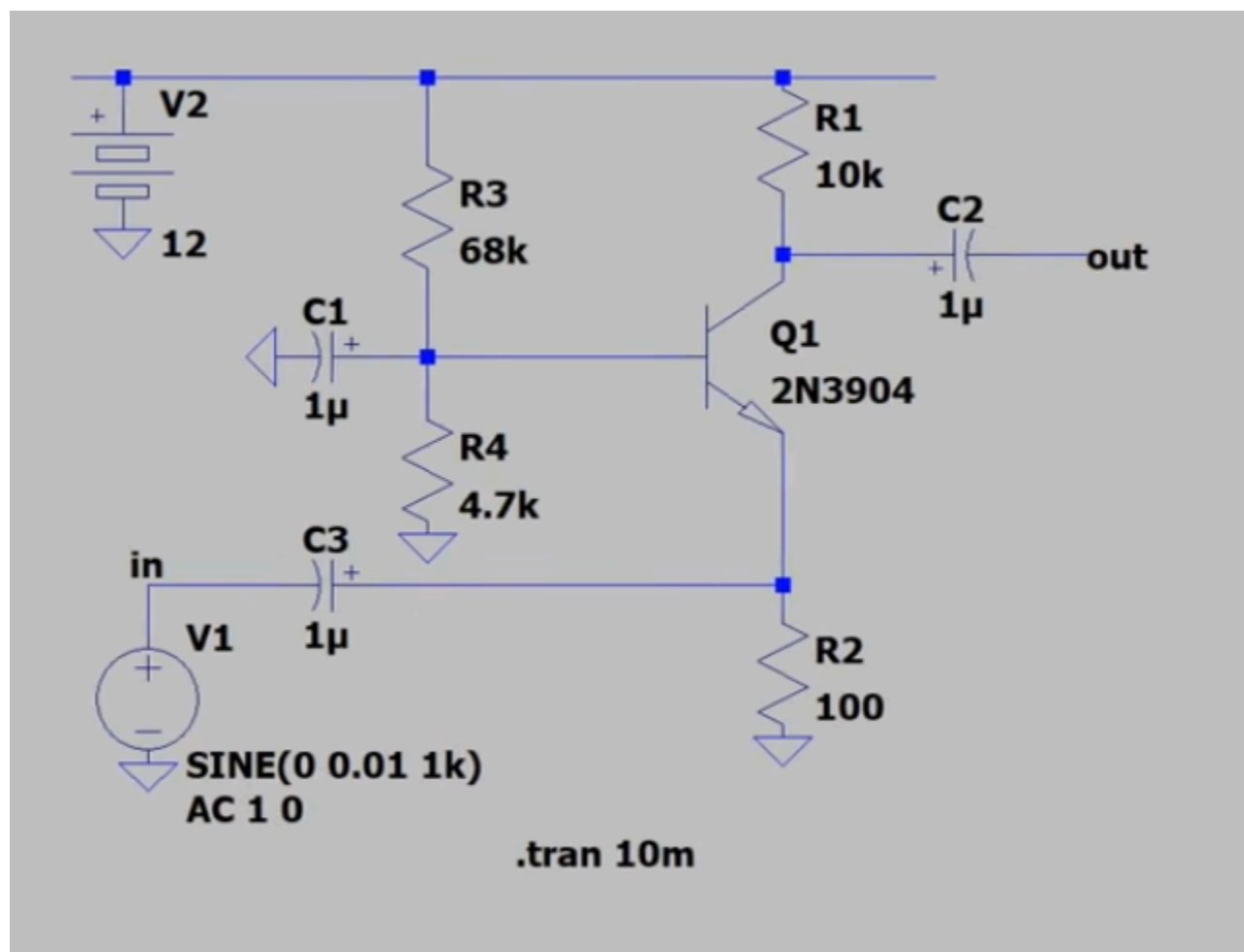
Aim:

To obtain the output waveform of CB Amplifier using LTSpice.

Apparatus Required:

LTSpice Tool

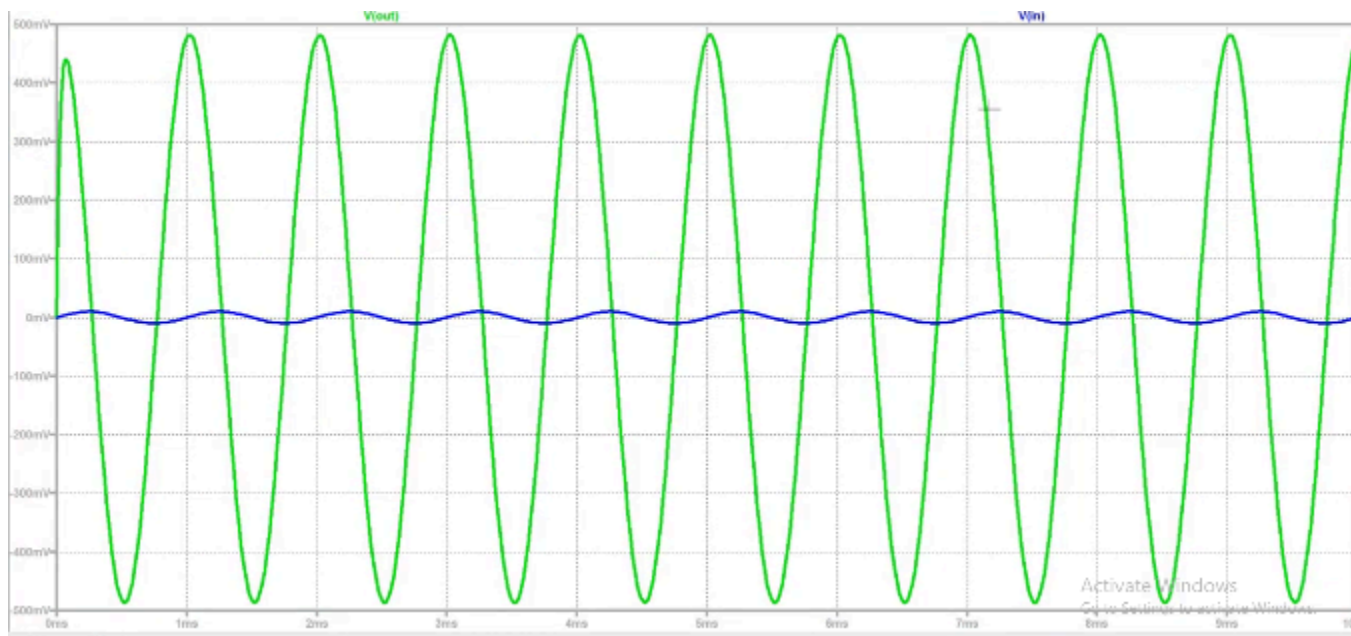
Circuit Diagram:



Procedure:

1. Connections are made as shown in the circuit diagram.
2. The DC power supply is switched ON.
3. The output waveform is displayed on the CRO.
4. The Amplitude and time period of the sine wave is noted.
5. The graph of output waveform is drawn.

Model Graph:



Result:

CB Amplifier circuit using LTSpice was completed and output obtained.

Case 6.1

Simulate the **frequency response** of a common-base amplifier using **BJT BC547** in LTSpice.

Given parameters:

- β (DC current gain) = 150
- $V_{BE} = 0.7$ V
- $V_{CC} = 12$ V

- $R_C = 3.3 \text{ k}\Omega$
- $R_E = 1 \text{ k}\Omega$
- $R_L = 2.2 \text{ k}\Omega$

Case 6.2

Simulate a **common-base amplifier** in LTSpice using **BC547** with the following conditions:

- $V_{CC} = 12 \text{ V}$, $V_{BE} = 0.7 \text{ V}$
- β varies from 100 to 300
- $R_C = 2.2 \text{ k}\Omega$, $R_E = 560 \Omega$
- Signal source = 10 mV AC

Case 6.3

Using LTSpice, design and simulate a common-base amplifier with BC547 ($\beta = 150$, $V_{BE} = 0.7 \text{ V}$).

Circuit parameters:

$V_{CC} = 12 \text{ V}$, $R_C = 4.7 \text{ k}\Omega$, $R_E = 1 \text{ k}\Omega$, $R_L = 3.3 \text{ k}\Omega$, and coupling capacitors = $10 \mu\text{F}$.

7. Simulation of Schmitt Trigger using LTSpice

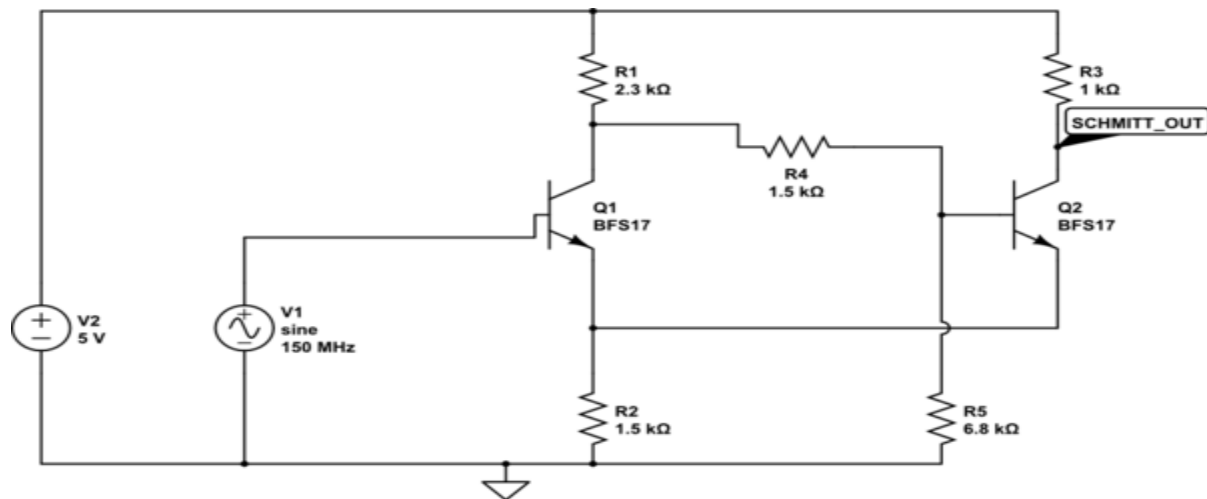
Aim:

To obtain the Output Waveform of Schmitt Trigger using LTSpice

Apparatus Required:

LTSpice Tool

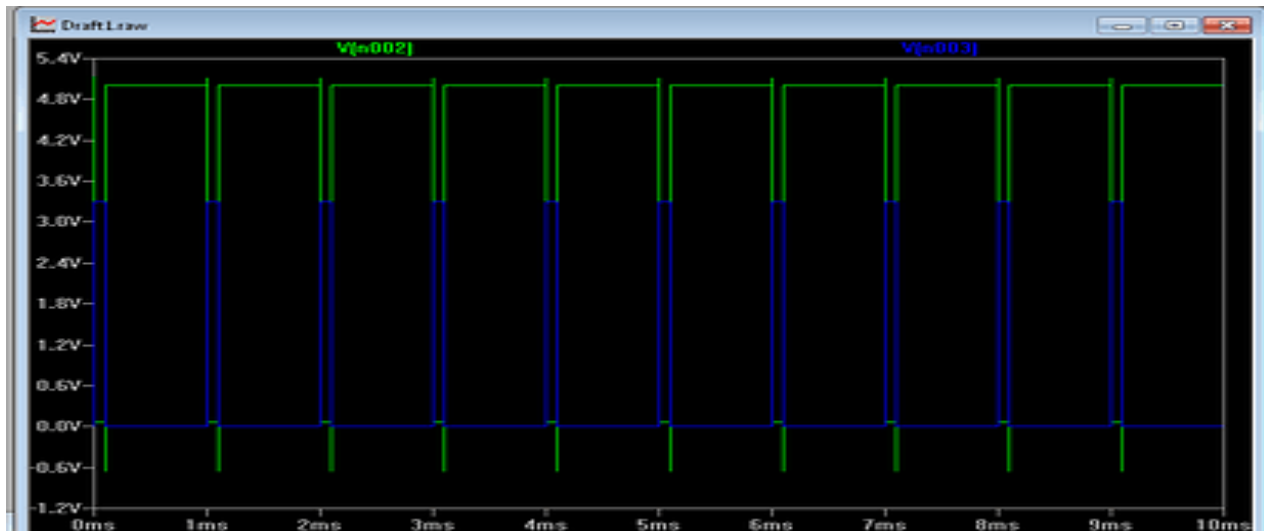
Circuit Diagram:



Procedure:

1. Connections are made as shown in the circuit diagram.
2. The DC power supply is switched ON.
3. The output waveform is displayed on the CRO.
4. The peak-to-peak Amplitude and time period of the sine wave is noted.
5. The graph of output waveform is drawn.

Model Graph:



Result:

Schmitt Trigger using LTSpice was completed and output obtained.

Case 7.1:

Design a SCHMITT TRIGGER using bipolar junction transistors (BJTs) by LTSPICE.

Case 7.2:

Design a SCHMITT TRIGGER using Op-amp 741 by LTSPICE.

8. Simulation of UJT Relaxation Oscillator using LTSpice

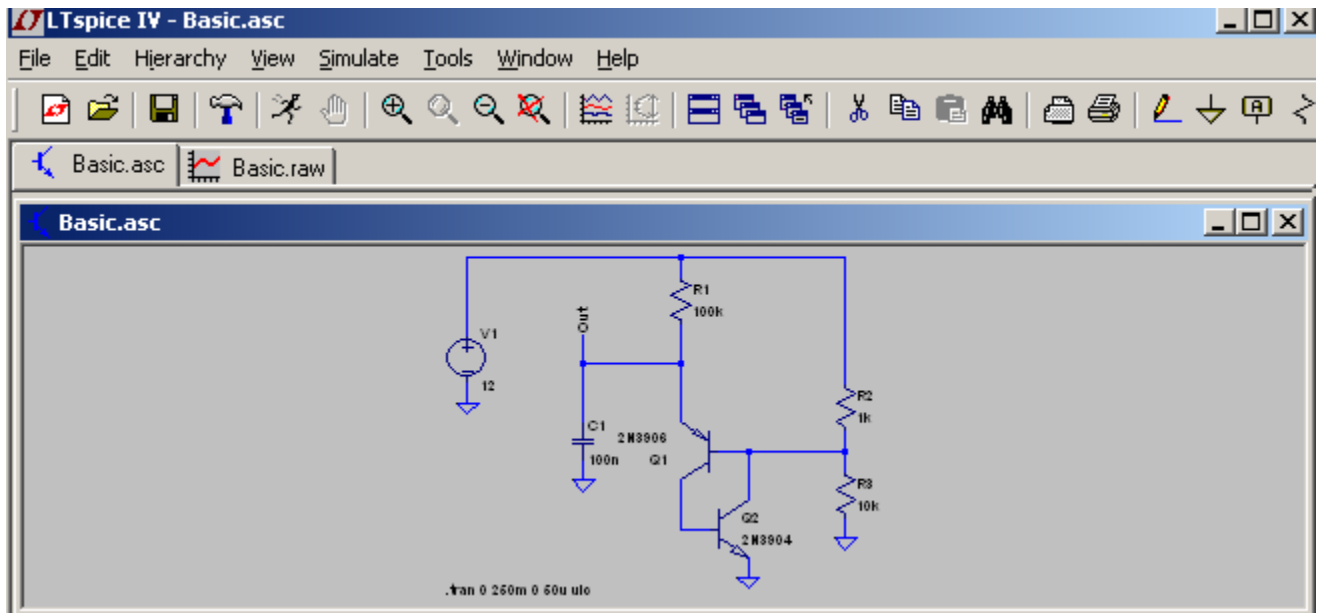
Aim:

To obtain the Output Waveform of UJT Relaxation Oscillator using LTSpice

ApparatusRequired:

LTspiceTool

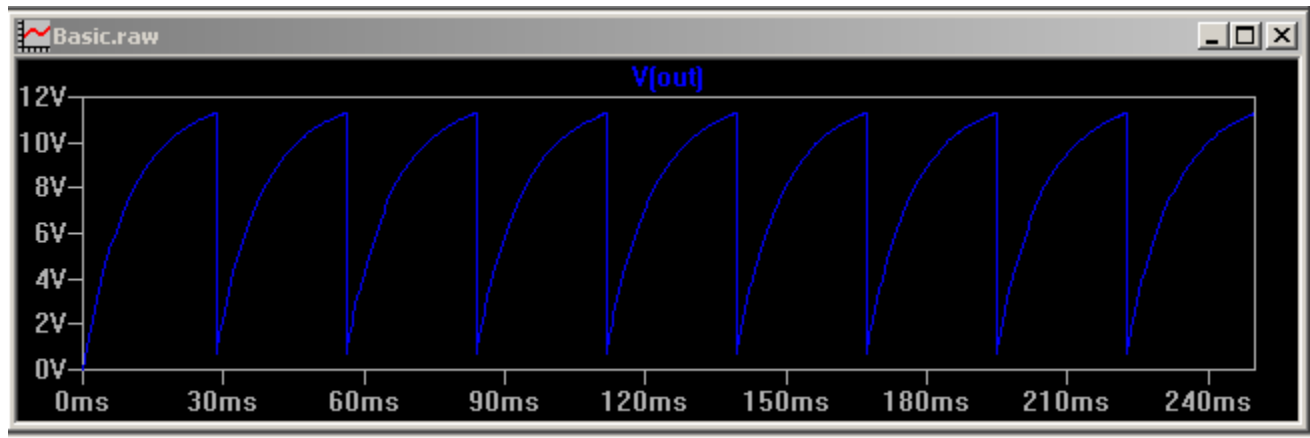
CircuitDiagram:



Procedure:

1. Connections are made as shown in the circuit diagram.
2. The DC power supply is switched ON.
3. The output waveform is displayed on the CRO.
4. The peak-to-peak Amplitude and time period of the sine wave is noted.
5. The graph of output waveform is drawn.

Model Graph:



Result:

UJT Relaxation Oscillator using LTSpice was completed and output obtained.

Case 8.1:

Design a UJT Relaxation Oscillator using bipolar junction transistors (BJTs) by LTSPICE.

Case 8.2:

Design a UJT Relaxation Oscillator using bipolar junction transistors with $C_1=10\text{nF}$ by LTSPICE.

9.Design of Hartley Oscillator Using EAGLE Software

Aim

To design a Hartley oscillator circuit and implement its PCB layout using EAGLE software, and to generate Gerber files for fabrication.

Apparatus / Software Required

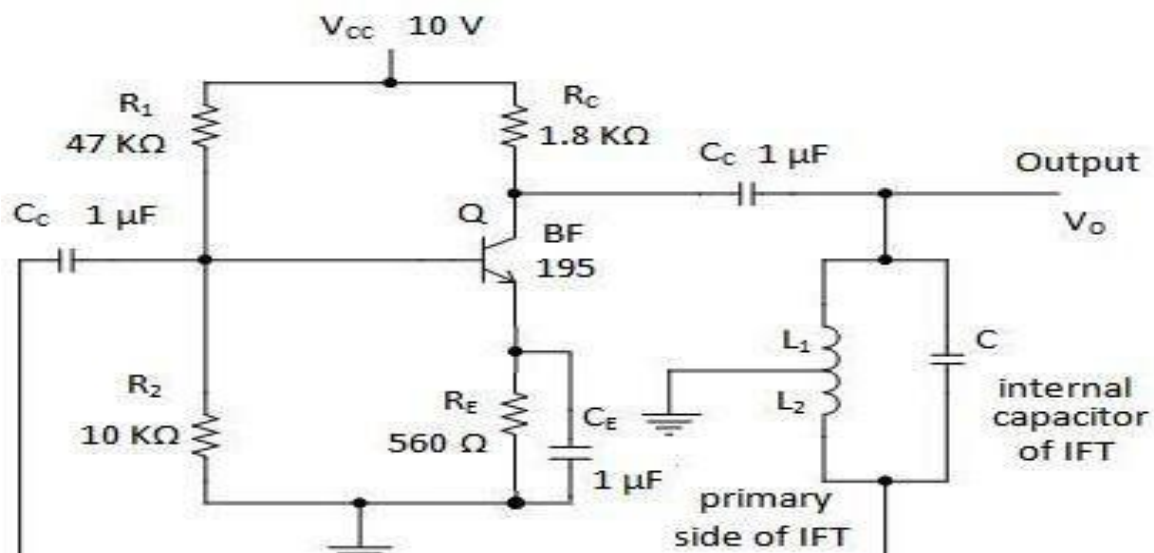
- EAGLE CAD Software
- Transistor
- Resistors, Capacitors, Inductors
- DC Power Supply (10 V)
- Gerber Viewer

Theory

A Hartley oscillator is an LC oscillator that uses a tapped inductor for feedback. The total inductance of the tank circuit determines the frequency of oscillation. Positive feedback is achieved through inductive coupling. The condition for sustained oscillations is given by the Barkhausen criterion.

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$$

Circuit Diagram



Procedure

Step 1: Schematic Design

- A new project is created in EAGLE.
- Components such as BC547, resistors, capacitors, and inductors are placed.
- Connections are made using the NET command.
- Electrical Rule Check (ERC) is performed.

Step 2: PCB Layout Design

- Schematic is converted to board layout.
- Board outline is defined.
- Components are placed to minimize noise and lead length.
- Traces are routed manually.
- Ground plane is created using POLYGON.

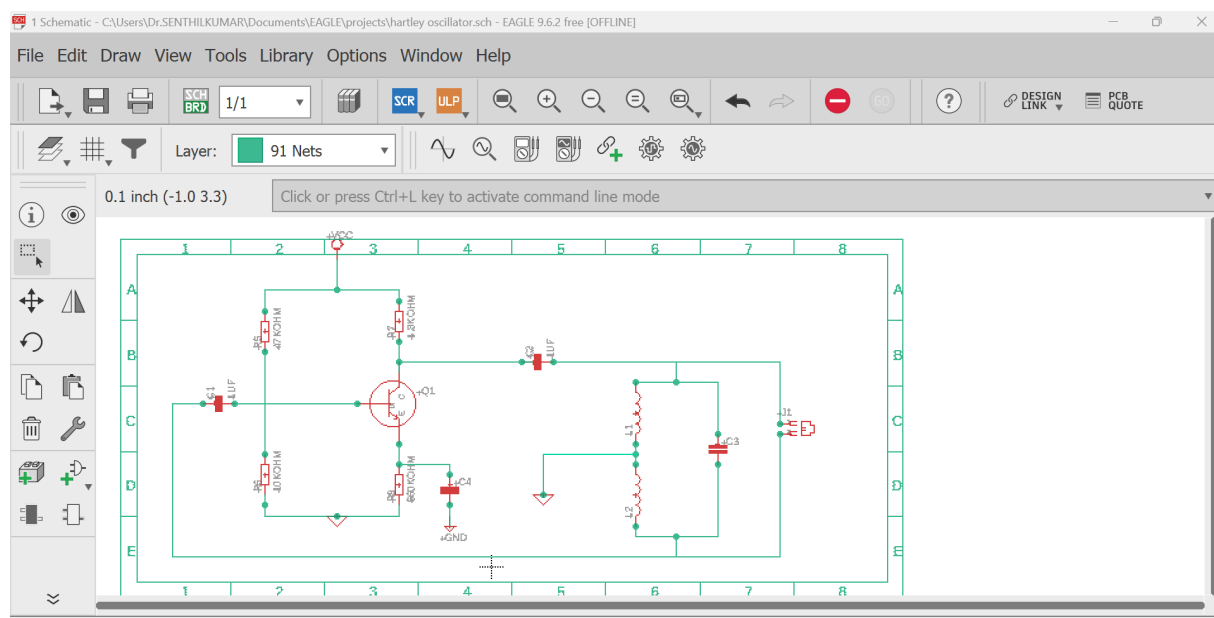
Step 3: Design Rule Check

- Track width and clearance are set.
- DRC is performed to remove layout errors.

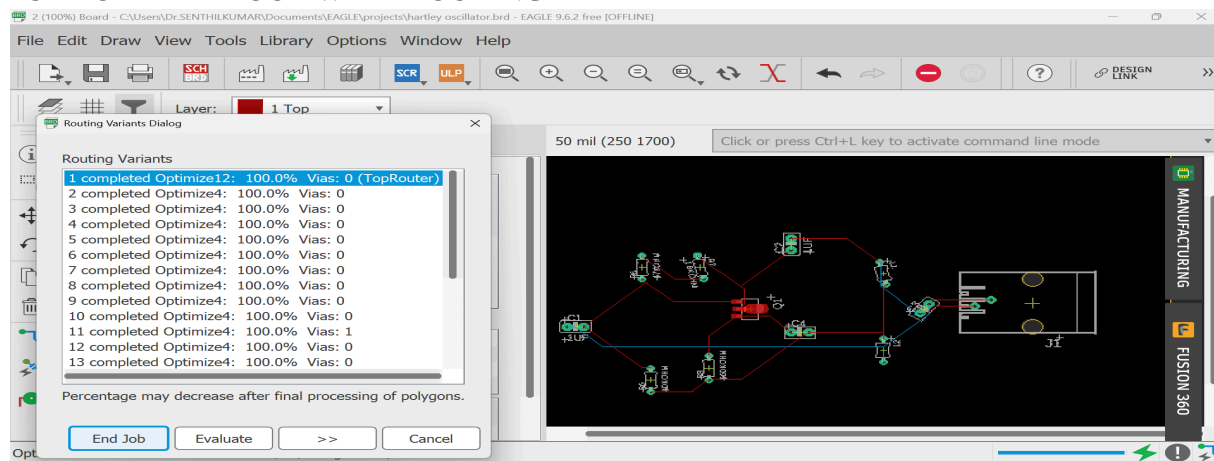
Step 4: Gerber File Generation

- CAM processor is used.
- Gerber and drill files are generated using RS-274X format.

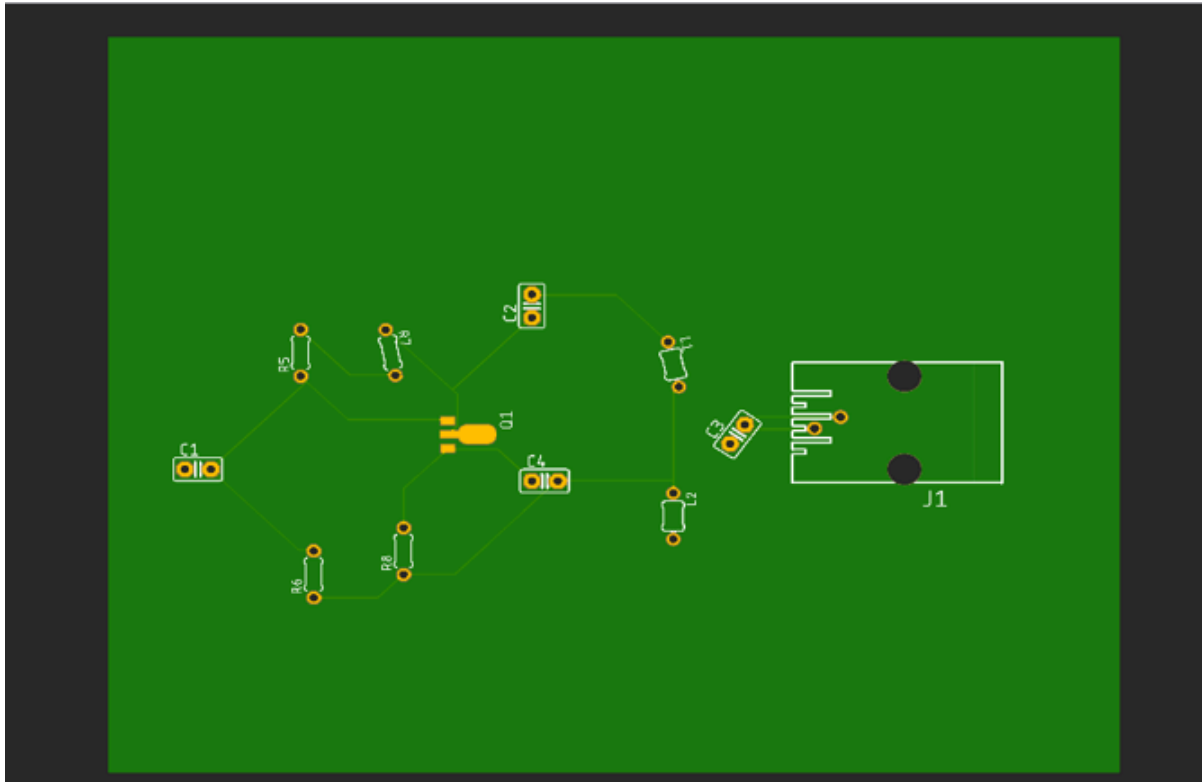
PCB SCHEMATIC DIAGRAM



PCB BOARD LAYOUT WITH ROUTING



MANUFACTUREING LAYOUT



GERBER FILE DETAILS

Result

The Hartley oscillator circuit was successfully designed using EAGLE software. The PCB layout was completed and Gerber files were generated for fabrication.

Test Case 1

- **Input Condition:** Apply DC supply $V_{CC}=5\text{ V}$
- **Design Values:** $L_1=10\mu\text{H}$, $L_2=10\mu\text{H}$, $C=1.2\text{Nf}$
- **USE SMT COMPONENTS ONLY**

Test Case 2:

- **Input Condition:** Apply DC supply $V_{CC}=15\text{ V}$
- **Design Values:** $L_1=20\mu\text{H}$, $L_2=20\mu\text{H}$, $C=20\text{ pf}$
- Transistor BC107
- **USE SMT AND THT COMPONENTS**

10.Design of Monostable Multivibrator Using EAGLE Software

Aim

To design a **monostable multivibrator circuit** and implement its **PCB layout using EAGLE CAD software**, and to generate **Gerber files** for PCB fabrication.

Apparatus / Software Required

- EAGLE CAD Software
- Transistor (BC547 / 2N2222) – 2 Nos
- Resistors and Capacitors (as per design)
- DC Power Supply (5 V / 9 V)
- Gerber Viewer

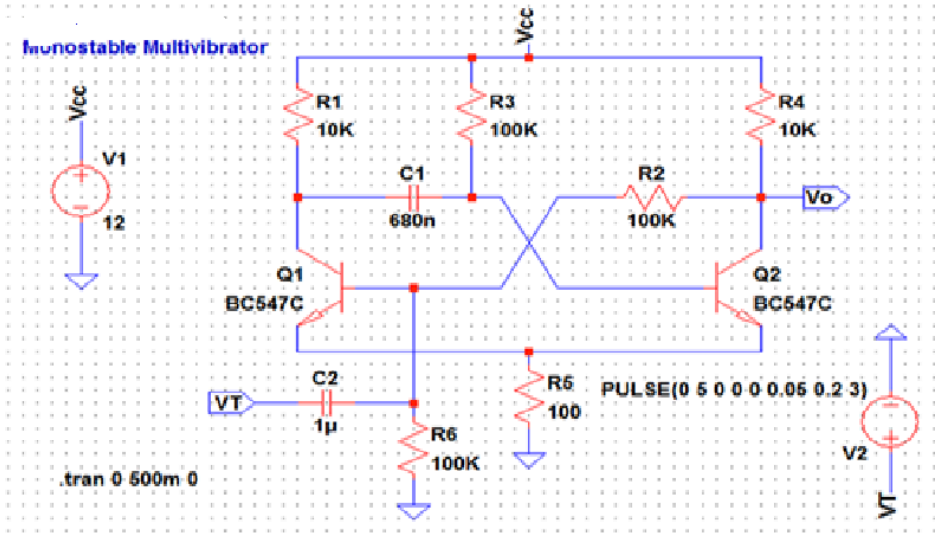
Theory

A monostable multivibrator has **one stable state and one quasi-stable state**. When a trigger pulse is applied, the circuit switches to the quasi-stable state for a fixed duration and then returns to the stable state automatically.

Pulse Width Equation

$T=0.69 \times R \times C_T = 0.69 \times R \times C_T=0.69 \times R \times C$

CIRCUIT DIAGRAM



Design Specifications (Sample Values)

Assumed Pulse Width

$T=1\text{ ms}$

Chosen Component Values

- $R = 1.5\text{ k}\Omega$
- $C = 1\text{ }\mu\text{F}$ Other Components

Component	Value
Transistors	BC547 (Q1, Q2)
RC (Collector Resistor)	4.7 kΩ
RB (Base Resistor)	47 kΩ
RE (Emitter Resistor)	1 kΩ
Coupling Capacitor	0.01 μF
VCC	9 V

Procedure in EAGLE

Procedure

Step 1: Schematic Design

- A new project is created in EAGLE.
- Components such as BC547, resistors, capacitors, and inductors are placed.
- Connections are made using the NET command.
- Electrical Rule Check (ERC) is performed.

Step 2: PCB Layout Design

- Schematic is converted to board layout.
- Board outline is defined.
- Components are placed to minimize noise and lead length.
- Traces are routed manually.
- Ground plane is created using POLYGON.

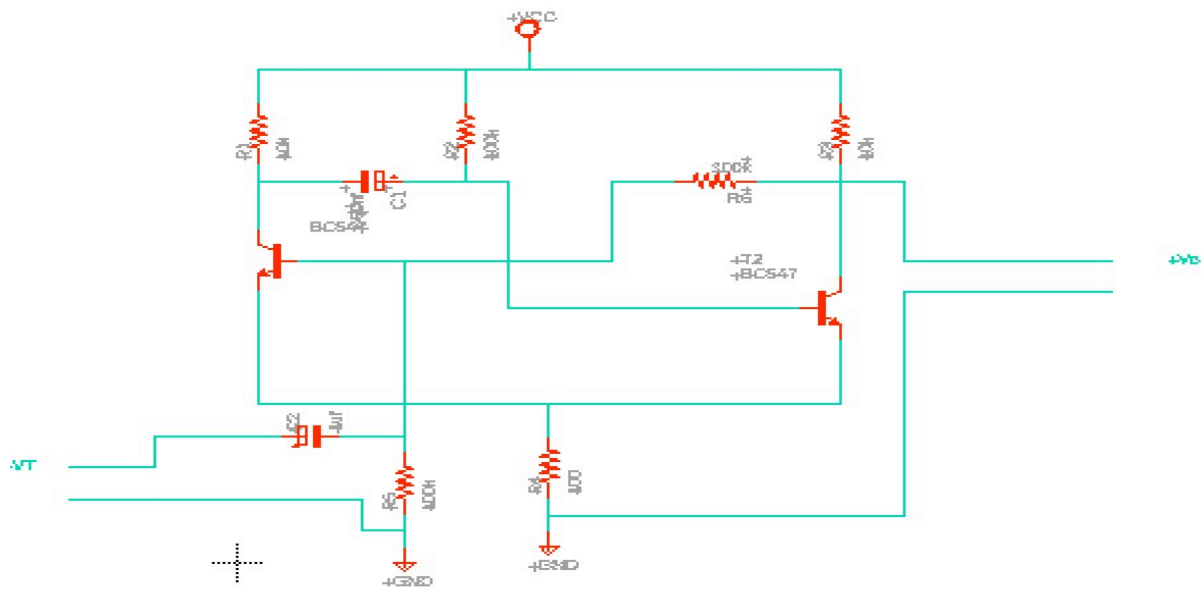
Step 3: Design Rule Check

- Track width and clearance are set.
- DRC is performed to remove layout errors.

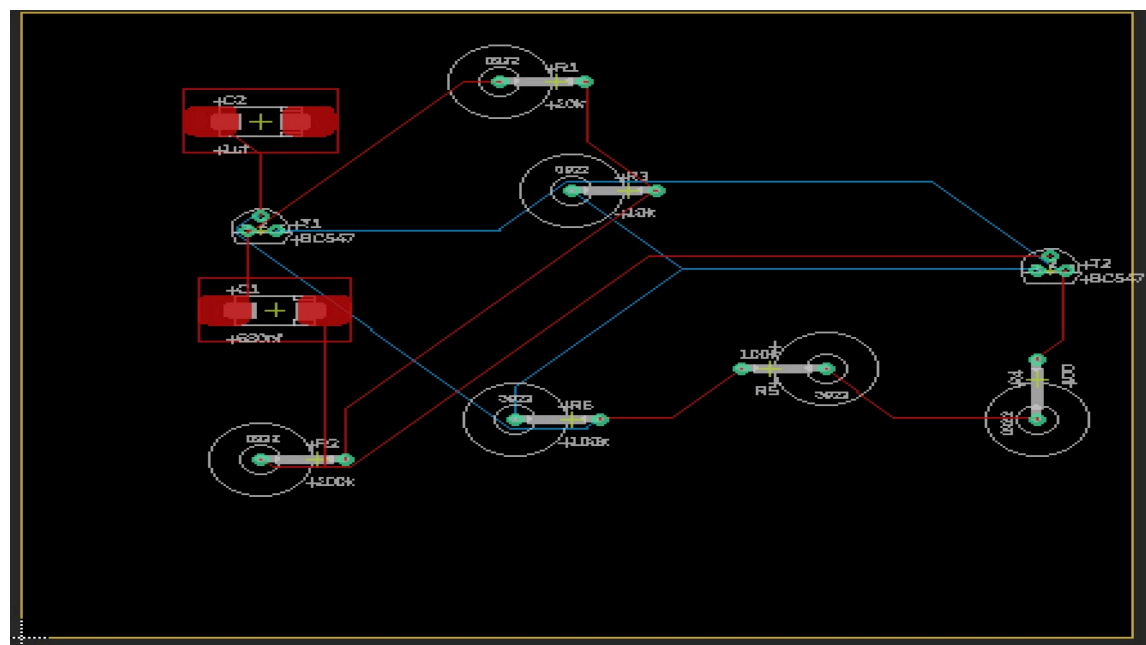
Step 4: Gerber File Generation

- CAM processor is used.
- Gerber and drill files are generated using RS-274X format.

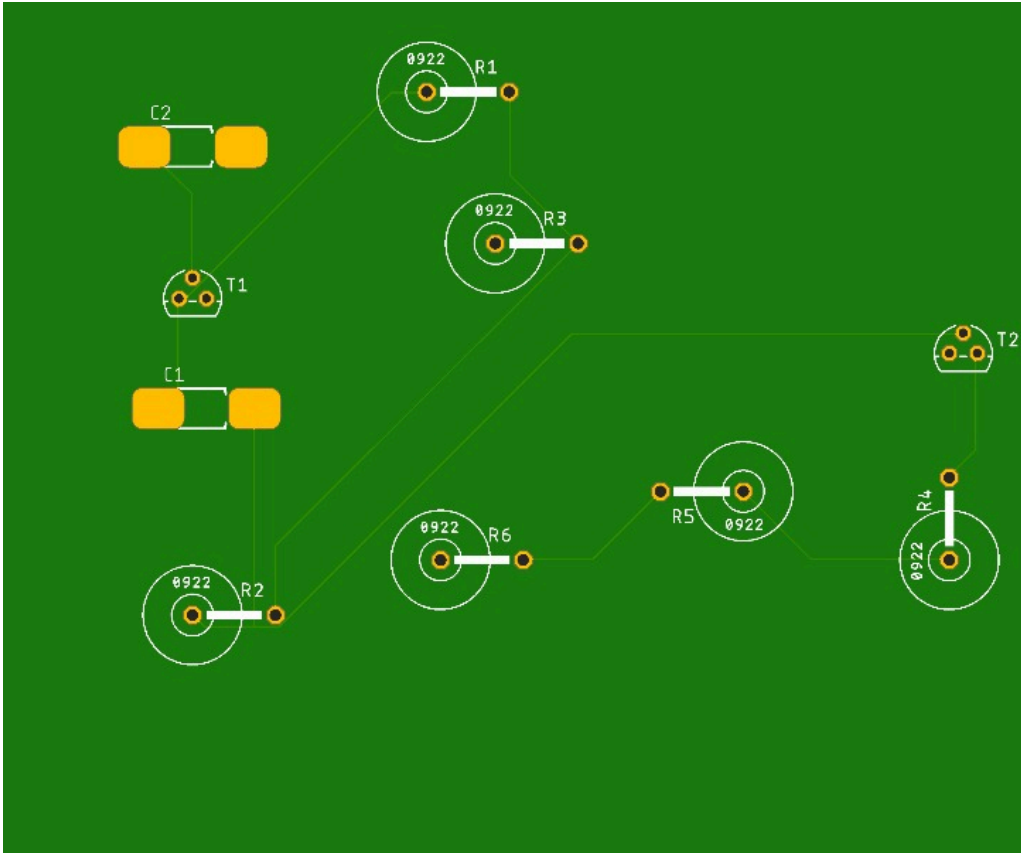
PCB SCHEMATIC DIAGRAM



PCB BOARD LAYOUT WITH ROUTING



MANUFACTUREING LAYOUT



GERBER FILE DETAILS

6. Result

The monostable multivibrator circuit was successfully designed using EAGLE software. The PCB layout was completed and Gerber files were generated for fabrication.

TEST CASE1:
Assumed Pulse Width
T=2 ms

Chosen Component Values

- **R = 2.5 kΩ**
- **C = 10 μF**
- **USE ONLY SMT COMPONENTS**

Other Components

Component	Value
Transistors	BC547 (Q1, Q2)
RC (Collector Resistor)	5.7 kΩ
RB (Base Resistor)	57kΩ
RE (Emitter Resistor)	1 kΩ
Coupling Capacitor	0.01 μF
VCC	15V

TEST CASE2:
Assumed Pulse Width
T=3 ms

Chosen Component Values

- **R = 3.5 kΩ**
- **C = 15μF**
- **USE SMT AND THT COMPONENTS**

Other Components

Component	Value
Transistors	BC547 (Q1, Q2)
RC (Collector Resistor)	7.7 kΩ
RB (Base Resistor)	27kΩ
RE (Emitter Resistor)	10 kΩ
Coupling Capacitor	0.01 μF
VCC	5V

